

Fundamentals of Robotics

Robotics

What Is a Robot?

A robot is a programmable machine that senses its environment, makes decisions based on that information, and acts on the physical world through motion or manipulation. Every robot, from a simple line-following cart to an industrial arm, is built from the same three functional blocks: sensing (gathering information about the world), processing (deciding what to do with that information), and actuation (turning decisions into physical motion). Understanding this sense-think-act loop is the foundation for everything else in robotics — every topic in this course is really about building or improving one of these three blocks.

Types of Robots

Robots are usually grouped by how they move and what they're built to do:

- Mobile robots — move through an environment on wheels, tracks, or legs (e.g. delivery robots, vacuum cleaners, rovers).
- Robotic arms / manipulators — fixed at a base, move an end effector through space to grip, weld, or assemble (common in factories).
- Humanoid robots — designed to mimic human form and movement, used in research and human-interaction settings.
- Drones (aerial robots) — fly using rotors or fixed wings, controlled by an onboard flight computer.
- Autonomous vs. teleoperated — autonomous robots decide their own actions from sensor data; teleoperated robots are directly controlled by a human operator.

Basic Mechanics: Degrees of Freedom

A robot's degrees of freedom (DOF) describe the number of independent ways it can move. A wheeled cart moving on a flat floor typically has 2-3 DOF (forward/back, turn, and sometimes strafe). A robotic arm used in manufacturing commonly has 6 DOF — enough to position and orient an object anywhere within its reach, the same number of independent motions a human arm and wrist provide. More DOF means more flexibility in movement, but also more motors, more computation, and more that can go wrong — good robot design finds the minimum DOF that gets the job done reliably.

Kinematics: Describing Motion

Kinematics is the study of motion without worrying about the forces that cause it — essentially the geometry of how a robot's joints relate to where its end effector (hand, tool, or wheel contact point) ends up in space.

- Forward kinematics: given the angles/positions of every joint, calculate where the end effector is. This is the easier direction — it's just applying geometry joint by joint.
- Inverse kinematics: given a target position for the end effector, calculate what each joint angle needs to be. This is harder — there can be multiple valid solutions, or none at all if the target is out of reach.
- Workspace: the full set of points a robot's end effector can physically reach, shaped by its link lengths and joint limits.

Center of Gravity and Stability

For any robot that has to stay upright — mobile robots, legged robots, drones on landing — the center of gravity (CG) matters as much as the mechanics. A robot is statically stable if its CG stays within its base of support (the area enclosed by its wheels or feet touching the ground) at all times. Robots with a narrow wheelbase or a high CG (like a tall robot with heavy components mounted up top) tip over more easily on turns or uneven ground. This is why real robot designs keep batteries and heavy motors low and centered whenever possible.

Robotics Safety Basics

Robots combine moving parts, electrical power, and often autonomous decision-making — all three are sources of real risk if ignored. Safe practice starts before you ever power on a robot:

- Always identify pinch points (where a joint or wheel can trap a hand or clothing) before running any motor for the first time.
- Keep a clear, immediate way to cut power — a physical emergency-stop switch, not just a software 'stop' command, since software can hang or lose connection.
- Test new code or mechanisms with the robot propped up off the ground (wheels spinning freely) before letting it move under its own power.
- Never work on a robot's electrical or motor system while it's powered — disconnect the battery first.
- For anything with exposed high torque (arms, grippers), assume it's strong enough to injure a hand and treat it accordingly, even at 'low' power settings.

Learn More

MIT OpenCourseWare — Introduction to Robotics

<https://ocw.mit.edu/>

GeeksforGeeks — Robotics Basics

<https://www.geeksforgeeks.org/robotics/introduction-to-robotics/>

freeCodeCamp YouTube Channel

<https://www.youtube.com/@freecodecamp>